

Junhuai Xu

Ph.D. Candidate, Department of Physics, Tsinghua University

Ph.D. Advisor: Zhigang Xiao

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Curriculum Vitae

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Profile

Ph.D. candidate in experimental nuclear and particle physics, working across detector instrumentation, scientific computing, and physics-informed reconstruction. I develop analysis workflows that connect detector calibration, response simulation, event reconstruction, inverse problems, statistical inference, and machine-learning methods for nuclear and neutrino experiments.

Research Experience

Nucleon Short-Range Correlations

Oct. 2023 – Present

CSHINE Experiment, RIBLL-1 (HIRFL), Lanzhou, China

- Extracted the SRC-induced high-momentum-tail fraction in ^{124}Sn , $R_{\text{HMT}} = (20 \pm 3)\%$, by comparing precision hard- γ spectra with detector-filtered IBUU-MDI transport calculations.
- Developed calibration and detector-response models for the CsI(Tl) γ -ray hodoscope used in bremsstrahlung measurements from low-energy heavy-ion collisions.
- Implemented Richardson–Lucy spectrum unfolding and quantified reconstruction/systematic uncertainties through pseudo-experiments, closure tests, and consistency checks.

Correlation Function Imaging

Feb. 2024 – Present

Department of Physics, Tsinghua University

- Developed a Richardson–Lucy imaging framework for femtoscopic two-particle correlations, reconstructing freeze-out source distributions without assuming a Gaussian source shape.
- Reconstructed identical non-Gaussian source functions for pp and $\bar{p}\bar{p}$ correlations in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV, providing coordinate-space evidence for matter–antimatter symmetry at freeze-out.
- Extended the imaging strategy toward three-dimensional source reconstruction with reproducible fit, resampling, and validation workflows for collaboration analyses.

Deep-Sea Neutrino Detection

Jun. 2024 – Present

TRIDENT Collaboration

- Built an atmospheric-neutrino simulation-to-reconstruction chain combining HONDA fluxes, FLUKA interactions, and Geant4 detector transport for a compact TRIDENT array.
- Developed graph-neural-network reconstruction of event class, direction, and energy from simulated detector responses, benchmarking performance for oscillation analyses.
- Evaluated atmospheric-neutrino oscillation-parameter sensitivity using Poisson-sampled pseudo-experiments and spectral comparisons across parameter hypotheses.

Experimental Experience

- **CSHINE Experiment** (Lanzhou, China, 2024): participated in beam-time data taking and high-energy γ -ray measurements with the CSHINE detector system.
- **SLEGS Beam Test at SSRF** (Shanghai, China, Jan. 2025): performed high-energy γ -ray response calibration of CsI(Tl) crystals using quasi-monochromatic photon beams and Geant4 simulations.
- **S π RIT Experiment** (RIKEN, Japan, 2024): supported detector operation and electronics control during SAMURAI beam time, contributing to stable data taking and run coordination.

Technical Skills

Physics analysis	Detector calibration, detector-response simulation, event reconstruction, spectrum unfolding, systematic uncertainty evaluation.
Inference	Inverse problems, Richardson–Lucy deconvolution/imaging, likelihood-based inference, pseudo-experiments, closure tests, resampling.
Machine learning Computing	Graph neural networks, PyTorch, physics-informed reconstruction, simulation-to-inference workflows. Python, C/C++, ROOT, Geant4, FLUKA-based workflows, Linux, Git, HPC/GPU-accelerated computing, LaTeX.

Education

Tsinghua University

Aug. 2022 – Jun. 2027 (Expected)

Ph.D. Candidate in Physics

Research areas: short-range correlations, heavy-ion collisions, femtoscopy, neutrino simulation, and machine-learning-based reconstruction.

South China Normal University

Aug. 2018 – Jun. 2022

Bachelor of Science

GPA: 4.06/5.00; Rank: 1/137. Thesis: *A Study of the Nuclear Medium Effect on Hadron Fragmentation Functions.*

Selected Publications

A complete publication list is available separately on my personal website.

- **Xu, J. et al.**, "Precise measurement of short-range correlations in nuclei from bremsstrahlung gamma-ray emission in low-energy heavy-ion collisions," *Phys. Rev. Res.* **7**, 043174 (2025).
- **Xu, J. et al.**, "Experimental study of bremsstrahlung γ -ray emission and short-range correlations in $^{124}\text{Sn}+^{124}\text{Sn}$ collisions at 25 MeV/nucleon," *Phys. Rev. C* **113**, 044613 (2026).
- **Xu, J. et al.**, "Reconstruction of Bremsstrahlung γ -rays spectrum in heavy ion reactions with Richardson-Lucy algorithm," *Phys. Lett. B* **857**, 139009 (2024).
- **Xu, J. et al.**, "Imaging Freeze-Out Sources and Extracting Strong Interaction Parameters in Relativistic Heavy-Ion Collisions," *Chin. Phys. Lett.* **42**, 031401 (2025).
- **Xu, J. et al.**, "Linear response of CsI(Tl) crystal to energetic photons below 20 MeV," *Nucl. Instrum. Meth. A* **1080**, 170787 (2025).

Awards

- First-class Comprehensive Scholarship, Tsinghua University (2025).
- Second-class Comprehensive Scholarship, Tsinghua University (2024).
- Outstanding Teaching Assistant, Tsinghua University (2024).
- Third Prize, National Undergraduate Mathematics Competition Final Round, Non-Math Major (2021).
- First Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM, 2020).